

EDA Project 1

Data Preparation

- ☐ Load the given dataset into a Pandas DataFrame.
- ☐ Provide a brief overview of the dataset, including:
 - Total number of rows.
 - Total number of columns.
- ☐ List all column names and identify their data types:
 - Numerical
 - Categorical
 - Datetime
- ☐ Identify the target variable (if one exists in the dataset).
- ☐ Explain what the dataset represents in 2–3 sentences.
- ☐ Describe why the dataset is suitable for an Exploratory Data Analysis (EDA) project.
- ☐ Summarize any notable characteristics of the dataset that may be useful for further analysis.

Data Cleaning

- ☐ Identify Missing Values
 - Check every column in the dataset for missing (null) values.
 - Calculate and report the percentage of missing data in each affected column.
 - Select an appropriate treatment method for each column and justify your choice:
 - Drop rows if only a small number of records are affected.
 - Replace missing numerical values with the mean or median, depending on the data distribution.
 - Replace missing categorical values with the mode (most frequent value).
 - Explain why the chosen method is suitable for preserving data quality.
- ☐ Detect and Remove Duplicate Records
 - Identify any duplicate rows in the dataset.
 - Remove all duplicate entries.
 - Report:
 - Number of duplicate rows found.
 - Number of duplicate rows removed.
 - New dataset shape (rows $\times$ columns) after duplicate removal.
- ☐ Detect and Handle Outliers
 - Select at least two numerical columns for outlier analysis.
 - Use the Interquartile Range (IQR) Method to detect outliers:
 - For each selected numerical column:
 - Report the lower and upper bounds.
 - State the number of outliers detected.
 - Explain the chosen treatment:
 - Remove the outliers,
 - Cap the extreme values, or
 - Retain them if they represent legitimate observations.
 - Justify the decision based on the dataset context.
- ☐ Summary of Data Cleaning
 - Provide a short paragraph summarizing all preprocessing activities, including:
 - How missing values were handled.

- Number of duplicate records removed.
- How outliers were treated.

Analysis and Insights

□ Single Column Analysis

- Pick one numerical column from the cleaned dataset.
- Create a histogram or box plot to show how the values are distributed.
- Write one simple observation about what the graph shows (e.g., spread, concentration, skewness, or outliers).
- Pick one categorical column from the dataset.
- Create a bar chart showing how often each category appears.
- Write one observation about which category is most common and which is least common.

□ Relationship Between Two Columns

- Create a correlation heatmap for all numerical columns using Seaborn.
- Find the two columns that are most strongly related (ignoring self-correlation).
- Report:
 - The names of both columns.
 - The correlation value.
 - A simple explanation of what this relationship means in real terms.
- Choose any two columns and show their relationship using:
 - A scatter plot, or
 - A grouped box plot, or
 - A table (crosstab).
- Write one observation about the pattern or trend you notice.

□ Looking at Multiple Columns Together

- Create a pair plot using at least three columns.
- Look at how the columns behave together in different combinations.
- Describe one interesting pattern or grouping that only becomes visible when looking at all selected columns together.
- Explain why this insight is useful for understanding the dataset.

□ Summary of Findings

- Write a short paragraph summarizing:
 - What you noticed in the single-column analysis.
 - The strongest relationship between two columns.
 - The key pattern found when looking at multiple columns together.
 - What these results tell you about the dataset overall.

Conclusion

□ Key Findings

- The distribution of the selected numerical variable showed with some extreme values, as seen in the histogram or other diagram indicating the presence of outliers.
- The correlation heatmap revealed a strong relationship between two numerical variables, suggesting one could help predict the other in a modelling context.
- The categorical bar chart showed an uneven distribution, where one category was significantly more frequent than others, indicating possible class imbalance.

□ Data Quality Assessment

- Some assumptions were made while handling missing values (using mean, median, or mode), which may slightly affect accuracy.
- Decisions made during outlier handling (removing or capping values) could influence future model performance.
- Overall, the dataset is cleaner but still carries minor risks from imputation and imbalance.

□ Possible Machine Learning Questions

- Can the target variable be predicted using the most strongly correlated features?
 - Because strong correlations suggest useful predictive relationships between variables.
- Can the dataset be used to classify records into categories based on numerical patterns?
 - Because visible patterns in distributions suggest separable groups.

□ Final Verdict

- The dataset is mostly ready for modelling, but additional steps like feature scaling, handling class imbalance, and validating preprocessing assumptions are recommended before building a machine learning model.

EDA Project 2

Choose any dataset of your interest from a public source such as Kaggle or other free source. It must have at least 500 rows and a mix of numerical and categorical columns. Perform a complete Exploratory Data Analysis covering data preparation, cleaning, analysis, and conclusion, following the same structure and requirements from the previous assignment. Justify every decision you make.